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Abstract—Coherent free-space optical links between satellites and optical ground stations must track a rapidly varying carrier phase induced by laser linewidth and Gamma-Gamma atmospheric turbulence, making carrier phase recovery a critical receiver function. Data-aided (DA) estimation achieves high accuracy at low SNR but pays a 1.25 dB pilot overhead, whereas non-data-aided (NDA) estimation wastes no bandwidth yet suffers from squaring loss that degrades its low-SNR accuracy. In this paper, we propose a per-block SNR-driven estimator switching scheme that selects, for each fading block, between a DA and an NDA carrier phase estimator based on the block’s measured SNR. The scheme selects the estimator with the lower bit error rate in 26 of 29 operating points, yielding a net SNR gain of up to about 1.9 dB in strong turbulence (naive, net of pilot overhead)¹.

I. INTRODUCTION

The growing deployment of low-Earth-orbit (LEO) satellite constellations has driven demand for high-capacity feeder links between satellites and optical ground stations (OGSs), targeting data rates of hundreds of Gbit/s and beyond [1]. Coherent free-space optical (FSO) communication is a leading candidate for such links, since methods using coherent detection provide better receiver sensitivity and enable higher-order modulation formats as compared to intensity-modulation direct-detection systems [1][UNRESOLVED CITATION: Paillier]. However, optical signal propagation through the atmosphere is affected by absorption, scattering, and refractive-index fluctuations—i.e., atmospheric turbulence—which cause rapid amplitude scintillation and a time-varying carrier phase that the receiver must track [2][UNRESOLVED CITATION: Johst]. Carrier phase recovery (CPR) is therefore an essential block at the coherent receiver, tasked with estimating and compensating for the residual phase noise introduced by the laser linewidth and the turbulent channel [UNRESOLVED CITATION: B11]. Because this phase noise evolves on the scale of the laser coherence time and the turbulence coherence time, the receiver must re-estimate the carrier phase for each fading block to maintain coherent demodulation; the choice of phase estimator—particularly whether it relies on embedded pilots—directly governs the residual phase error and the achievable bit error rate.

Two established CPR strategies trade off estimation accuracy against spectral efficiency. Data-aided (DA) estimation, which divides out modulation using known pilot symbols,

¹All reported gains are net of the pilot power penalty of 1.25 dB ($10 \log_{10}(4/3)$); the bit error rate is computed over data symbols (768 bits per block) unless otherwise stated.

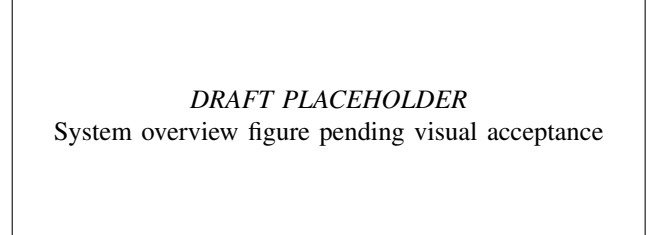


Fig. 1. System overview placeholder for the coherent FSO link and per-block estimator switching.

achieves high estimation accuracy at low signal-to-noise ratio (SNR) but pays a bandwidth penalty for the inserted pilots—on the order of $10 \log_{10}(4/3) = 1.25$ dB at a one-in-four pilot spacing [UNRESOLVED CITATION: Shieh-Djordjevic 2010]. Non-data-aided (NDA) estimation, in contrast, removes modulation by raising the received samples to the M_0 -th power (the Viterbi-and-Viterbi estimator), wastes no bandwidth on pilots, but suffers from squaring loss that degrades its low-SNR phase estimation accuracy [3]. The two estimators therefore dominate in disjoint SNR regions: DA is preferable where the signal is weak, and NDA where the signal is strong.

In this paper, we propose a per-block SNR-driven estimator switching scheme that selects, for each fading block, between a data-aided (DA) and a non-data-aided (NDA) carrier phase estimator based on the block’s measured SNR. The scheme selects the estimator with the lower bit error rate in 26 of 29 operating points, yielding a net SNR gain of up to about 1.9 dB in strong turbulence (naive, net of pilot overhead)². The remainder of this paper details the channel and signal model (Section II), the DA/NDA estimators and the switching rule (Section III), and simulation results over Gamma-Gamma fading channels under downlink and uplink conditions (Section IV).

II. SYSTEM MODEL

As depicted in Fig. 1, we consider a coherent satellite-to-ground (and uplink) FSO link operating with (8, 8)-16APSK modulation, where the constellation comprises $M_0 = 8$ points per ring [UNRESOLVED CITATION: B11]. The atmospheric channel is modeled as a Gamma-Gamma block-fading channel [2]: the channel undergoes block fading (quasi-static), such that the fading coefficient h_b remains constant over a block

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of $N_{\text{blk}} = 100$ symbols and varies independently between blocks. The Gamma-Gamma shape parameters (α, β) are set to (4.0, 3.0), (2.5, 1.8), and (1.5, 0.8) for the weak, moderate, and strong downlink turbulence regimes, respectively, and (1.2, 0.9) and (1.0, 0.7) for the moderate and strong uplink regimes. The pair (α, β) determines the Rytov variance σ_R^2 , a standard measure of turbulence strength [2]; smaller values of α and β correspond to stronger turbulence and, consequently, deeper amplitude fading.

The discrete-time complex baseband received signal at symbol index k within block b is modeled as

$$r_k = h_b s_k e^{j\theta} + n_k,$$

where s_k is the transmitted (8, 8)-16APSK symbol, θ denotes the carrier phase offset (denoted ϕ in [UNRESOLVED CITATION: B11]; we follow the θ notation of [3]), and n_k is complex AWGN with one-sided power spectral density N_0 ; the signal-to-noise ratio is $\gamma = E_s/N_0$, where E_s is the per-symbol energy. The block length $N_{\text{blk}} = 100$ symbols is chosen well below the coherence time of both the laser phase noise and the atmospheric fading, which justifies the quasi-static (block-fading) assumption [UNRESOLVED CITATION: Johst]. The frame structure inserts one pilot symbol every four symbols, yielding a pilot spacing of 1/4 and a pilot overhead of 25%, equivalent to a 1.25 dB SNR penalty $10 \log_{10}(4/3)$ [UNRESOLVED CITATION: Shieh-Djordjevic 2010]. This pilot density follows the frame structure adopted in [UNRESOLVED CITATION: B11], in which one pilot symbol is interleaved every three data symbols; the periodic arrangement lets the DA estimator sample the carrier phase at regular intervals across the block, while the three-to-one data-to-pilot ratio keeps the overhead moderate. The residual carrier phase is modeled as a Wiener phase noise process—i.e., a random walk with independent zero-mean Gaussian increments—of per-symbol variance

$$\sigma_\theta^2 = 2\pi \Delta\nu T_s \quad (\text{= } \sigma_p^2 \text{ in B11}),$$

[UNRESOLVED CITATION: B11] where $\Delta\nu = 10$ kHz is the laser linewidth and T_s is the symbol period. Carrier phase recovery is performed per DSP block of $N_{\text{DFT}} = 256$ samples, and the HD-FEC threshold is 3.8×10^{-3} (7% overhead) [UNRESOLVED CITATION: B11].

III. METHOD

We consider two carrier phase estimators that operate independently on each fading block, differing only in how they remove the modulation from the received samples. The data-aided (DA) estimator removes modulation by dividing each received pilot sample by the known pilot symbol p , averages the resulting phases over the N_p pilot symbols within the block, and takes the argument:

$$\hat{\theta}_{\text{DA}} = \angle(r_p \cdot p^*). \quad (1)$$

DRAFT FIGURE PLACEHOLDER
Carrier phase recovery and per-block estimator
switching mechanism pending

Fig. 2. Draft placeholder for the carrier phase recovery and per-block estimator-switching mechanism.

The non-data-aided (NDA) estimator instead raises every sample in the block to the M_0 -th power, averages, and scales the resulting argument by $1/M_0$:

$$\hat{\theta}_{\text{NDA}} = \frac{1}{M_0} \angle \left(\sum_k r_k^{M_0} \right). \quad (2)$$

where $M_0 = 8$ for (8, 8)-16APSK (one period per ring). The DA estimator achieves high accuracy at low SNR since the pilot symbol is known exactly at the receiver, but spends 25% of the bandwidth on pilots (1.25 dB overhead); the NDA estimator wastes no bandwidth yet suffers from squaring loss that degrades its phase-estimation accuracy at low SNR [3]. The two estimators therefore dominate in disjoint SNR regions. A practical difference is that the DA estimate is modulation-format independent, since the pilot symbols are known a priori and carry no payload, whereas the NDA estimate's reliability is governed by the M th-power squaring loss, whose severity scales with M_0 : for the (8, 8)-16APSK constellation with $M_0 = 8$, the noise amplification incurred in raising to the eighth power is considerably larger than for lower-order formats such as QPSK ($M_0 = 4$), which narrows the NDA estimator's usable SNR range and is precisely why a per-block selection between the two is beneficial on this constellation.

To exploit this complementarity, the receiver measures the per-block SNR

$$\gamma_{\text{blk}} = |h_b|^2 E_s / N_0, \quad (3)$$

i.e., the SNR over a single fading block of N_{blk} symbols within which the channel coefficient h_b is constant, and selects the estimator with the lower expected bit error rate: select $\hat{\theta}_{\text{DA}}$ when $\gamma_{\text{blk}} < \gamma_{\text{th}}$, and $\hat{\theta}_{\text{NDA}}$ otherwise. Here h_b denotes the per-block channel amplitude, estimated from the received block, and γ_{th} is the fixed effective-SNR switching threshold. The BER curves exhibit an observed, regime-dependent crossover that motivates this choice, but the threshold is not re-estimated from that crossing. The scheme thus performs a per-block, SNR-driven selection between the DA and NDA estimators rather than introducing a new estimator or a new closed-loop component; it adapts an existing DA/NDA choice to the locally prevailing SNR, and the fraction of operating points in which this selection recovers the lower-BER estimator is quantified in Section IV-B.

Because the two estimators dominate in disjoint SNR regions, committing to either one alone forgoes the advantage

the other provides outside its region. An all-DA receiver pays the 1.25 dB pilot overhead at every SNR, including the high-SNR region where the NDA estimator—unburdened by pilots—already achieves the lower BER; conversely, an all-NDA receiver incurs the squaring-loss penalty at low SNR, where the DA estimator’s pilot-based estimate is more accurate [3]. The per-block switching rule recovers the lower-BER choice in each region rather than paying either penalty globally. Regarding the switching criterion, the per-block SNR γ_{blk} is a natural statistic for this selection because the DA/NDA trade-off is itself governed by SNR: γ_{blk} is available at the receiver without additional overhead (it is computed from the already-received block), whereas a smoothed or genie-aided SNR estimate would either introduce latency or require side information unavailable in practice. A further distinction is that the DA estimate is independent of the payload constellation, since the pilot symbol is known a priori, whereas the NDA estimator must raise the received samples to the M_0 -th power matched to the constellation geometry ($M_0 = 8$ for (8,8)-16APSK); the switching scheme thus inherits the modulation-format independence of the DA branch in the low-SNR region where the DA estimator is selected.

The threshold γ_{th} is set to a fixed effective-SNR value, chosen to separate the low-SNR region where the DA estimator yields the lower BER from the high-SNR region where the NDA estimator does. This choice involves a trade-off: setting γ_{th} too low leaves a low-SNR region in which the NDA estimator is erroneously selected and suffers from squaring loss, whereas setting it too high retains the DA estimator—and its pilot overhead—in a high-SNR region where the NDA estimator is already superior. Aligning γ_{th} with this region boundary balances these two error regions. In terms of implementation cost, the scheme requires only one per-block SNR estimate (γ_{blk} , a ratio of known quantities) and one comparison against γ_{th} per block, on top of the two phase estimates that any single-estimator receiver would compute; this incurs negligible additional complexity relative to a fixed DA or fixed NDA receiver.

IV. RESULTS

The performance of the DA and NDA estimators, and of the proposed switching scheme, is evaluated over the Gamma-Gamma block-fading channel of Section II via Monte Carlo simulation. For each operating point, the simulation runs over $N_{\text{blocks}} = 400$ fading blocks per seed (yielding at least 10^5 bits) and averages over 30 independent seeds for the benchmark configuration to capture the randomness of the turbulent channel. The SNR is swept from 5 to 20 dB for the AWGN reference and from 5 to 26 dB for the turbulent regimes, covering both the low-SNR region where the DA estimator is preferable and the high-SNR region where the NDA estimator dominates. Six regimes are evaluated in total—weak, moderate, and strong downlink turbulence; moderate and strong uplink turbulence; and an AWGN (no-fading) reference—spanning the range from benign to severely faded channels. The AWGN reference serves as an implicit baseline

TABLE I
DRAFT STRUCTURE PLACEHOLDER FOR THE NET SNR-GAIN
COMPARISON. NUMERICAL ENTRIES AND THE FAIR/NAIVE REPORTING
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Regime	Reference	Measure	Value
Strong downlink	Fixed NDA	HD-FEC point	TBD
Moderate uplink	Fixed NDA	HD-FEC point	TBD
Strong uplink	Fixed NDA	HD-FEC point	TBD

against which the effect of turbulence is isolated. The principal metric is the bit error rate P_b evaluated at the 7% HD-FEC threshold of 3.8×10^{-3} , at which the net SNR gain of the switching scheme is reported.

A. Impact of Turbulence on DA and NDA Performance

Figure 3 reports the bit error rate P_b versus SNR for the DA and NDA estimators across the six considered regimes (weak, moderate, and strong downlink turbulence; moderate and strong uplink turbulence; and the AWGN reference), with BER curves plotted down to the minimum reliably estimable level for each regime. The AWGN reference, included as an implicit baseline, shows the largest gap between the two estimators in favor of NDA, since the absence of fading removes the DA estimator’s low-SNR advantage. As turbulence strengthens, the DA and NDA curves intersect: the two estimators are each preferable in a distinct SNR region, and the crossover SNR moves to lower values as turbulence grows, decreasing from 17.9 dB (weak) through 16.8 dB (moderate) to 10.7 dB (strong). Below γ_{th} the DA estimator yields the lower BER; above it the NDA estimator does. Two qualitative trends are visible in the data. First, the gap between the two estimators’ curves is largest in the AWGN reference, where the absence of fading leaves the DA estimator’s pilot overhead uncompensated, and narrows as turbulence strengthens. Second, the crossover SNR moves to lower values as turbulence grows (from 17.9 dB through 16.8 dB to 10.7 dB), so the SNR window in which the DA estimator is preferable widens with turbulence strength. This crossover structure is precisely what the per-block switching rule exploits.

B. Gain of the Switching Scheme

Table I summarizes the net SNR gain of the proposed scheme relative to a fixed NDA estimator, measured at the 7% HD-FEC threshold of 3.8×10^{-3} .

Two regimes are distinguished. *Low-SNR avoidance*: in the low-SNR region ($\gamma_{\text{blk}} < \gamma_{\text{th}}$), selecting the DA estimator avoids the NDA estimator’s squaring-loss-driven degradation, yielding a net SNR gain of 1.3–2.3 dB over a fixed NDA estimator. *Strong-turbulence net gain*: in the strong-turbulence and uplink regimes the scheme attains a net SNR gain of about 1.2–1.9 dB (naive, net of pilot overhead), specifically about 1.3 dB (downlink strong), about 1.2 dB (uplink moderate), and

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about 1.9 dB (uplink strong). Across all 29 operating points, the scheme selects the lower-BER estimator in 26 of the 29 points, recovering the gain that either estimator used alone would forgo whenever the two estimators disagree.

V. CONCLUSION

In this paper, we proposed a per-block SNR-driven estimator switching scheme that selects, for each fading block of a Gamma-Gamma satellite-to-ground channel, between a data-aided (DA) and a non-data-aided (NDA) carrier phase estimator. In 26 of 29 operating points this selection coincides with the estimator that achieves the lower bit error rate, yielding a net SNR gain of up to about 1.9 dB in strong turbulence (naive, net of pilot overhead)⁴. Future work is underway to strengthen the switching criterion, to extend the evaluation to additional turbulence regimes, and to validate the scheme on uplink experimental data.

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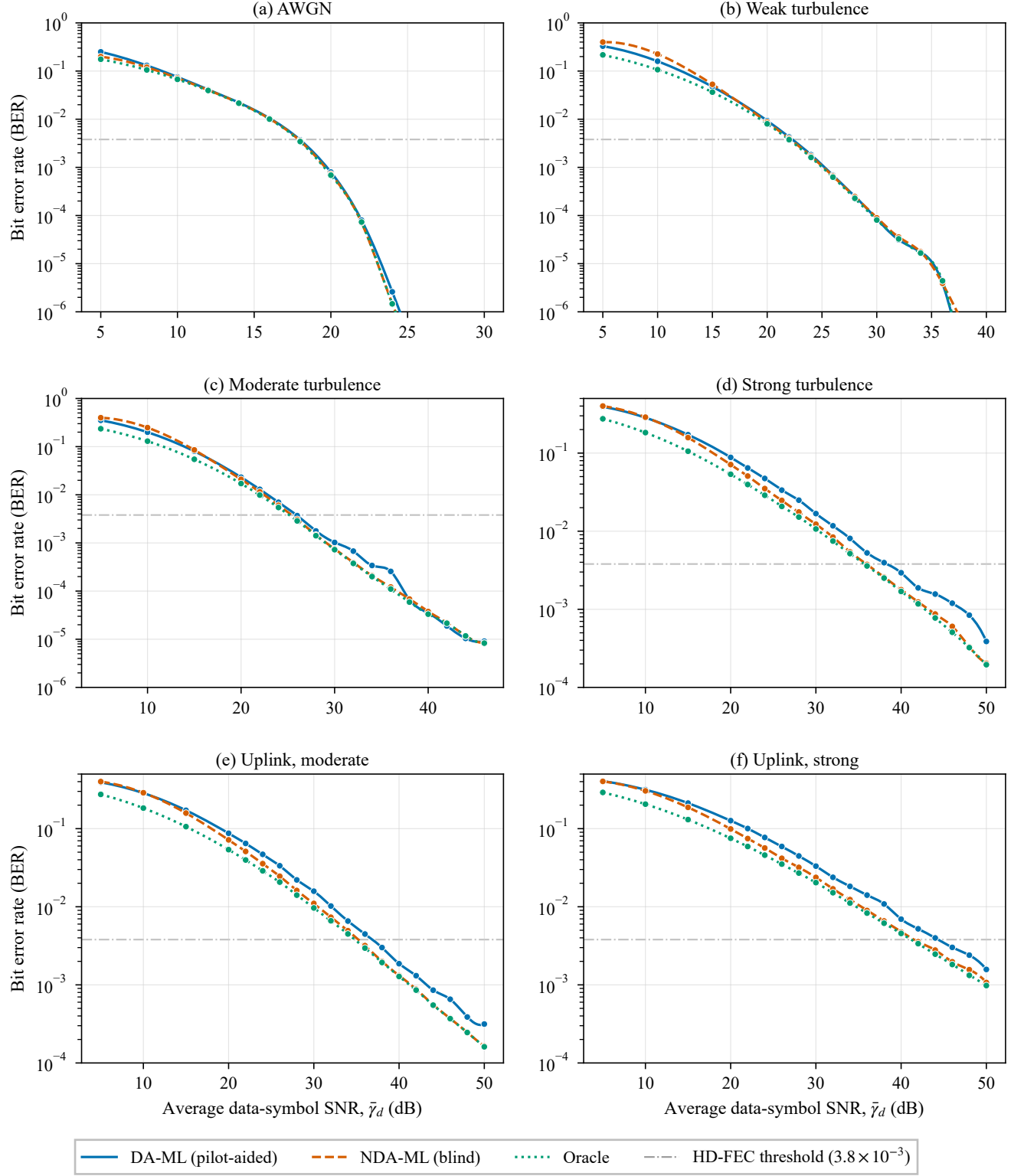


Fig. 3. Bit error rate P_b versus SNR for the DA and NDA estimators across the considered channel regimes.

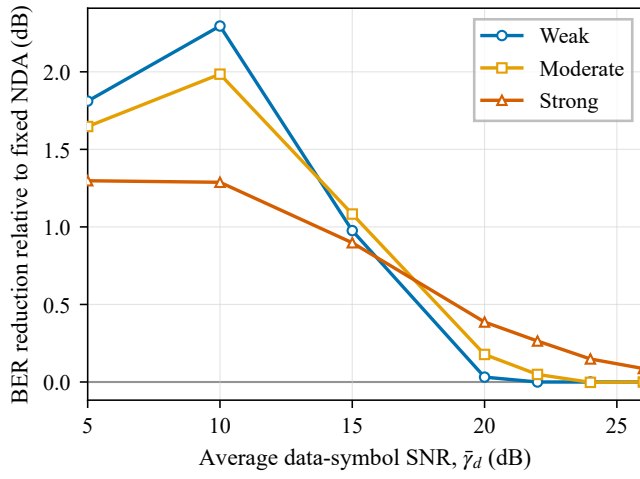


Fig. 4. Net SNR gain of the switching scheme relative to a fixed NDA estimator at the HD-FEC reference point.

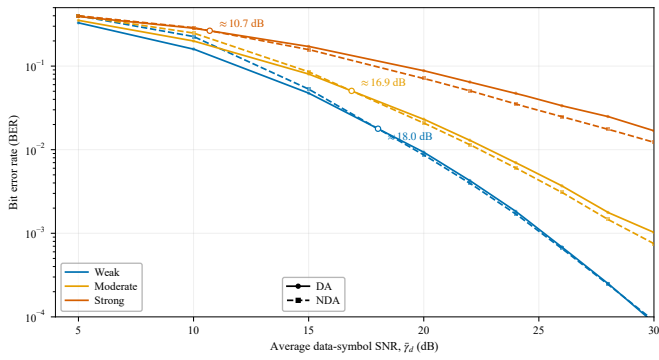


Fig. 5. Crossover SNR at which the DA and NDA BER curves intersect across turbulence regimes.